

Pranav Elavarthi

rzg8qh@virginia.edu • 571-294-4114 • Herndon, Virginia

Education

University of Virginia | Expected Graduation May 2027

BS in Computer Science and Applied Mathematics Minor | GPA: 4.00 | Dean's List

- Relevant Coursework: Data Structures and Algorithms 1 & 2, Computer Systems and Organization, Software Development Essentials, Introduction to Cybersecurity, Discrete Math and Theory

Thomas Jefferson High School for Science and Technology | Graduated June 2024

GPA: 4.584 | SAT: 1570/1600 | National Merit Scholar | AP Scholar with Distinction

Experience

GuidePoint Security | Cybersecurity Engineer Intern | September 2025 - December 2025

- Implemented a Python CLI tool to automate end-to-end SOC environment deployment, orchestrating VM provisioning via Ludus/Proxmox and triggering Ansible playbooks for tool installation and configuration
- Integrated and automated core SOC tooling including Splunk (SIEM), Velociraptor (EDR), Tenable/OpenVAS (vulnerability scanning), Wireshark, and Splunk Forwarder across segmented VLANs
- Reduced SOC lab deployment time from 2–6 weeks to ~30 minutes, enabling ~99.6% faster environment setup

Amp Health | Software Engineer Intern | June 2025 - August 2025

- Developed mobile application using Ionic/Angular framework, implementing real-time computer vision with Google MediaPipe for biomechanical movement tracking, supporting both iOS and Android
- Built responsive UI with Ionic components including camera integration, real-time canvas overlays, progress indicators, and touch-optimized controls while maintaining 30+ FPS performance
- Replaced third-party SDK costing \$8,000/month with custom MediaPipe solution

Obscurity Labs | Software Engineer Intern | July 2023 - August 2023

- Developed a CLI and API tool using Python and Unix in order to extract file metadata from Windows VHDs
- Created a fully functioning webpage that allows users to interact and access metadata and hash information
- Created a REST API that allows for file metadata lookups and pulls information from MongoDB database

Fairfax Collegiate | Assistant Instructor | July 2022 - August 2022

- Taught a class on Python game development to 20 middle school students
- Covered basic programming and software development skills, data structures, and pygames library
- Prepared lecture slides, assisted debugging, and walked through live coding demos

Projects

Augmented Reality Drone Control

- Developed an intuitive AR drone control system where users can control drone movements via hand gestures
- Utilized Microsoft Mixed Reality Toolkit HandPhysicsService and Unity game engine for frontend
- Built back end of the application using a Flask web server and djitellopy Python library
- Presented findings at Thomas Jefferson Symposium to Advance Research (TJSTAR)

Inter-Variability Study of COVLIAS 1.0

- Analysis of different Machine Learning models ability to predict COVID severity based on CT scans of lungs
- Co-authored paper published in journal Diagnostics: <https://pubmed.ncbi.nlm.nih.gov/34829372/>

Skills

- **Programming Languages:** Python, Java, Javascript, Typescript, C, R, x86 assembly, HTML/CSS
- **Tools:** Docker, MongoDB, MySQL, Angular, Tableau, Unity, Git, AWS, MATLAB
- **Other Technical Skills:** Agile, Test-Driven Development Augmented Reality, Mobile Development